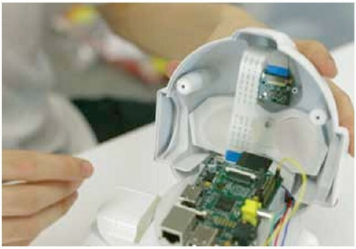
An open source approach to robotics

http://www.rapiro.com/

Shota Ishiwatari, a Japanese robot maker, has come up with Rapiro, a low-cost, customisable, programmable Raspberry Pi-based humanoid robot that can be put to personal use or for science and technology learning in schools. Rapiro, according to the maker, costs just a quarter of current aesthetic robot kits, and one-tenth the price of current Linux-powered humanoid robot kits. Rapiro has 12 servo motors, which enable it to walk, turn around the waist and neck, and handle even fine movements like gripping a pen. But all you need is a screwdriver and a little time to assemble it.

Rapiro is voice-activated. It can be controlled by the owner's voice, or with a mobile phone or gaming handset. It can also be connected to the Internet, and notify you of emails, Facebook messages, etc. It can even be programmed to safeguard your home or water your plants when you are away!



The Open Twist: Rapiro comes with a total of 12 specially-designed servo motors, one for its neck, one in the waist, four for the two feet, and the final six for its two arms. The torque of the six servos in the neck, waist and two feet, is 2.5 kgf-cm. The torque of the six servos in the two arms is 1.5 kgf-cm. Both have operating speeds of 0.12 sec/60°. Both have maximum angles of 180°. The servo control board is programmable and is completely compatible with the Arduino. Technically, Rapiro can work with or without a Raspberry Pi, but it is the Pi that adds all the josh to it, so it is better to fit one in! The design allows the Raspberry Pi and camera module to be mounted on the head. PSD distance sensors and speakers can also be mounted. Once Shota Ishiwatari's Kickstarter goal is achieved, there are plans to publish the sample code and 3D data openly for community customisation.

Sixth Sense

http://www.pranavmistry.com/projects/sixthsense/

Pranav Mistry's SixthSense is an interesting wearable gesture computing technology, which involves converting natural hand movements into digital information, to interact

with computer systems. The SixthSense prototype comprises a pocket projector, a mirror and a camera. The hardware components are coupled in a pendant-like device, and connected to the mobile computing device in the user's pocket. The projector projects visual information enabling surfaces, walls and physical objects around us to be used as interfaces; while the camera recognises and tracks the user's hand gestures and physical objects using computer-vision based techniques. The software program processes the video stream data captured by the camera, and tracks the locations and movements of the coloured markers at the tip of the user's fingers. This information is interpreted into gestures, which in turn act as instructions to an application. SixthSense supports multi-touch and multi-user interaction.

The Open Twist: SixthSense is open source. Its hardware specifications and software code are available at http://code.google.com/p/sixthsense/ licensed under the General Public License (GPL) v3. It is evident from the forum that people are interested in the project for learning and developmental purposes. They are also helping make it better.

SixthSense can also be used for commercial purposes with permission and appropriate licences for the end product. Mistry, an MIT Media Lab researcher, explained in an earlier *EFY Times* interview: "I come from India, an area where till a few years back the notion of technological advances has always been associated with the western world; to advances aimed at making the life of the western world better and better. But if you observe, life in the western world is good already and we need to break this model. It is the two-thirds other world that needs the technological advances so that the life of people in these countries becomes better. While I could have made more money if I had sold the technology to a big company, I believe I will get more blessings if I share the technology out in the open for the benefit of the masses." END

By: Janani Gopalakrishnan Vikram

The author is a technically-qualified freelance writer, editor and hands-on mom based in Chennai.